ASTANA IT UNIVERSITY

Laboratory Report

Sensors for IoT

Precision Agriculture Monitoring System

"AquaGuard"

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1. Abstract

This report describes the development and testing of AquaGuard, an IoT-based Precision Agriculture Monitoring System. The prototype integrates soil moisture, temperature, humidity, and illuminance sensors on an ESP32 microcontroller. Sensor data is transmitted via MQTT to a ThingsBoard cloud dashboard. An automated rule-based irrigation algorithm activates a water pump when soil moisture exceeds the dry threshold and suppresses irrigation when rain is forecast, achieving water-efficient automated crop management. Testing in the Wokwi simulation environment confirmed correct behaviour across all test scenarios.

2. Introduction

Traditional irrigation relies on fixed schedules rather than real-time soil conditions, leading to overwatering and water waste -a critical concern in arid regions such as Kazakhstan. Precision agriculture, enabled by IoT sensor networks, allows data-driven irrigation decisions that align water delivery with actual crop demand (Ahmed et al., 2018).

This project implements a low-cost ESP32-based monitoring station that collects multi-parameter environmental data, visualises it on a ThingsBoard dashboard, and applies an automated irrigation logic including a predictive rain-forecast feature. The prototype directly supports Kazakhstan's agricultural digitalisation goals (Baydildina et al., 2020).

Objectives

- Monitor soil moisture at two simulated depths using capacitive sensor proxies.
- Measure ambient temperature, humidity, and illuminance.
- Transmit telemetry to ThingsBoard via MQTT over Wi-Fi.
- Implement and validate an automated irrigation decision algorithm.
- Evaluate system behaviour under simulated field conditions.

3. System Architecture

The system is organised into three layers: Data Acquisition (sensors), Edge Processing (ESP32), and Cloud Visualisation (ThingsBoard). Sensor data is sampled

every 3 seconds, with MQTT telemetry published every 5 seconds. Critical events trigger Telegram Bot notifications via HTTPS.

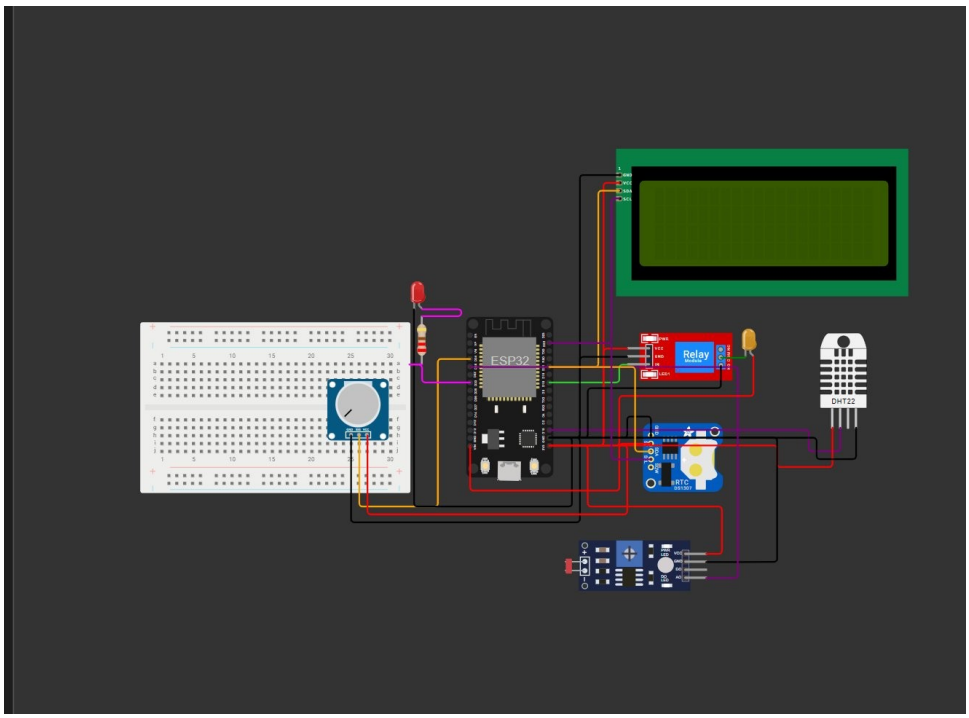

Figure 1. AquaGuard Wokwi simulation circuit -ESP32 connected to DHT22, potentiometer (soil sensor), LDR, relay, LCD2004, DS1307 RTC, LED, and push button

4. Hardware Implementation

Table 1 summarises all hardware components used in the prototype.

Table 1. Bill of Materials

Component	Model	Pin	Purpose
Microcontroller	ESP32 DevKit V1	—	Processing, Wi-Fi, MQTT
Soil moisture ×2	Capacitive / pot	D34, D35	Volumetric water content
Temp & humidity	DHT22	D15	Air temperature and RH
Light sensor	LDR (BH1750 target)	D35	Illuminance measurement
RTC module	DS1307	D21/D22	Telemetry timestamps
Relay module	5 V single channel	D18	Controls water pump
LCD display	LCD2004 I2C 20×4	D21/D22	Local status readout
Rain forecast button	Tactile push button	D4	Simulates rain input

Sensor Selection: Capacitive vs. Resistive

Capacitive soil moisture sensors measure the dielectric constant of soil rather than passing current through metal probes. This eliminates **electrolytic corrosion** -the dominant failure mode of resistive sensors -and produces consistent output in saline soils typical of irrigated fields. In this prototype, potentiometers simulate the analogue voltage output of capacitive probes. In field deployment, FC-28 or FDR probes would be used (Zhang et al., 2010).

5. Software and Data Processing

The firmware is written in C++ (Arduino framework) and runs on the ESP32. Key libraries include PubSubClient for MQTT, LiquidCrystal_I2C for the LCD, DHT for sensor reading, RTClib for timestamps, and HTTPClient for Telegram alerts. Telemetry is serialised as a JSON object containing eight keys: temperature, humidity, soil_moisture, light_lux, pump_active, led_active, rain_forecast, and rain_prob, and published to ThingsBoard topic v1/devices/me/telemetry every 5 seconds (Ahmed et al., 2018).

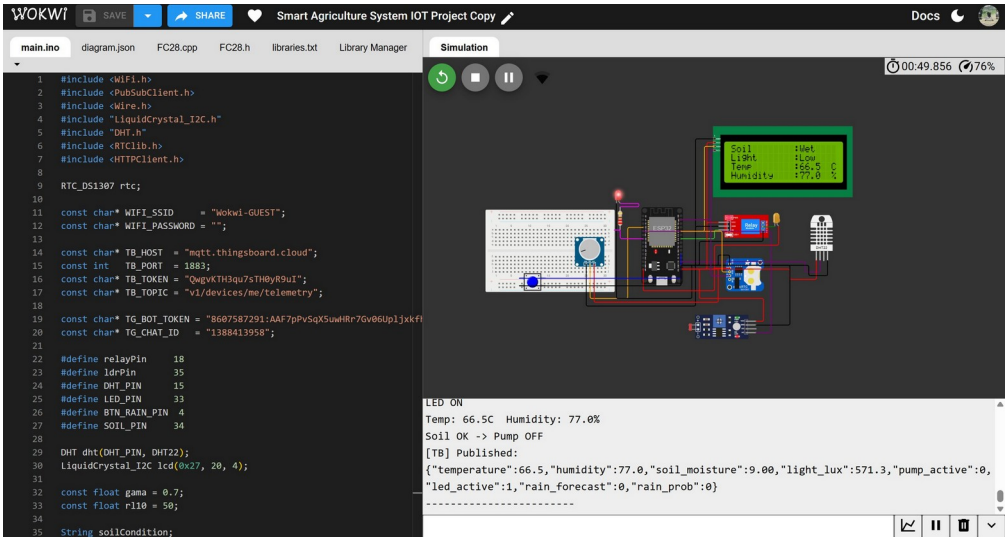

Figure 2. Wokwi normal monitoring state -LCD showing Soil: Wet, Light: Low, Temp: 66.5 °C, Humidity: 77.0%. Serial monitor confirms [TB] Published telemetry.

ThingsBoard Dashboard

The ThingsBoard dashboard displays four analogue gauge widgets (temperature, humidity, soil moisture, light), a time-series line chart, a latest-telemetry attributes card, and pump/LED status cards. Data appears in real time as MQTT messages are received.

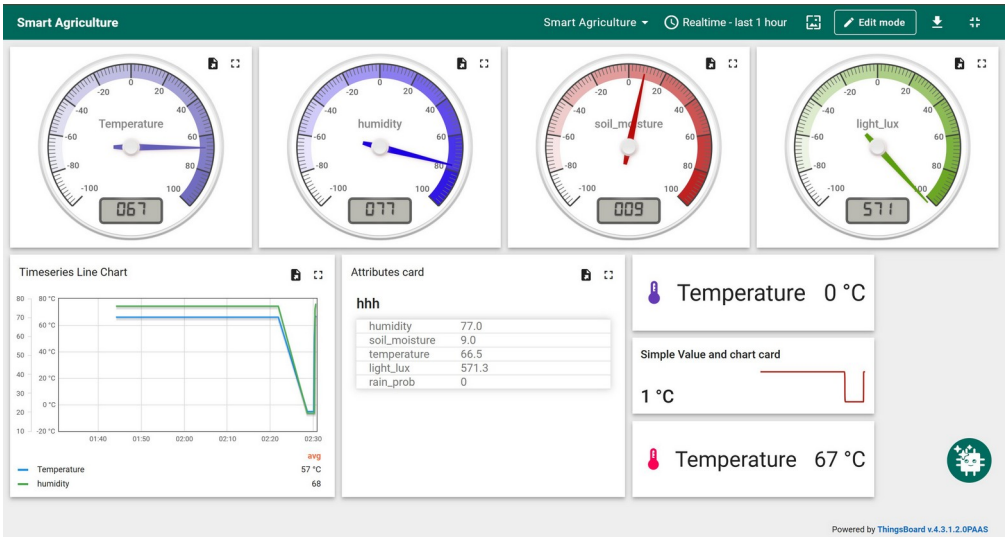

Figure 3. ThingsBoard dashboard showing gauges (Temperature 67, Humidity 77, Soil 9, Light 571), time-series chart, and attributes table with all telemetry keys confirmed.

6. Decision-Making Algorithm

The irrigation logic evaluates soil moisture state and rain forecast on every loop cycle:

- **Irrigation ON:** soil_moisture > 66.66% (Dry) AND rain_forecast = FALSE → relay HIGH, Telegram 'IRRIGATION STARTED'.
- **Predictive Skip:** soil_moisture > 66.66% AND rain_forecast = TRUE → relay LOW, Telegram 'PREDICTIVE SKIP -saving water'.
- **No Action:** soil_moisture ≤ 66.66% (Wet/Normal) → relay LOW.
- **Grow-Light:** light_lux < 807 → LED ON; else LED OFF.

```
IF soil_moisture > 66.66 THEN
    IF rain_forecast THEN pump = OFF  // predictive skip
    ELSE pump = ON                    // irrigate now
ELSE pump = OFF
IF light_lux < 807 THEN led = ON ELSE led = OFF
```

7. Testing and Results

Testing was conducted in Wokwi, which faithfully emulates the ESP32's ADC, GPIO, I2C, and Wi-Fi interfaces. The DHT22 model was set to 66 °C / 76% RH. The soil moisture potentiometer was manually adjusted between Wet, Normal, and Dry states to validate algorithm branches.

Irrigation Activation Test

When the potentiometer was set to maximum (100% dry), the pump activated immediately. The LCD displayed 'Soil is DRY! Pumping water...' and a Telegram

notification 'IRRIGATION STARTED -Soil: 82.0% (Dry)' was dispatched and confirmed with '[TG] Sent OK'.

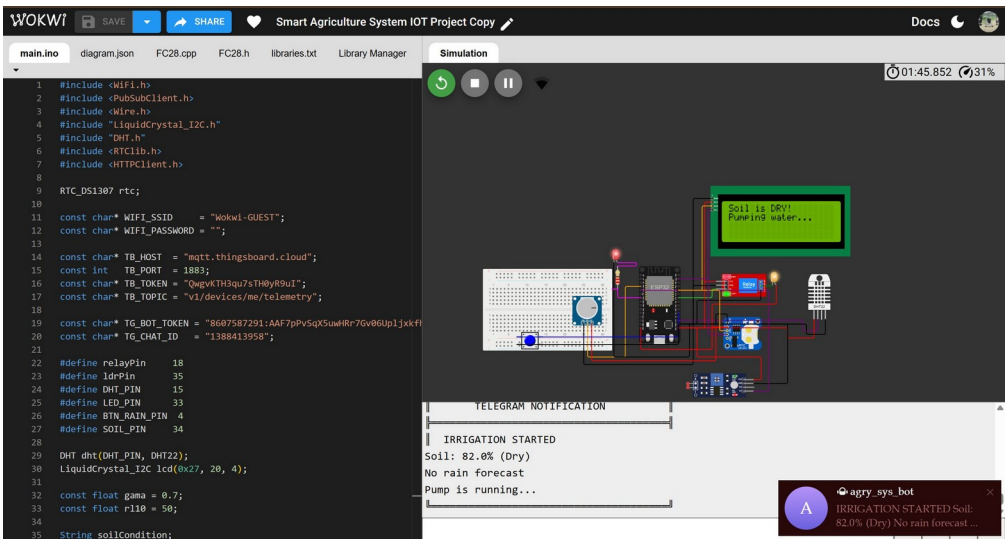

Figure 4. Irrigation activation: LCD shows 'Soil is DRY! Pumping water...', serial monitor displays IRRIGATION STARTED notification, Telegram bot alert visible bottom-right.

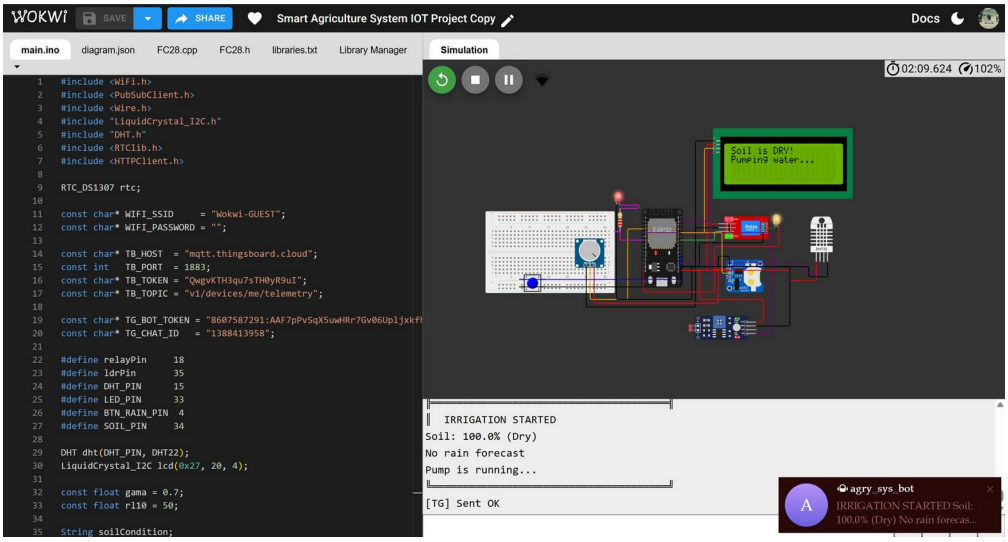

Figure 5. Telegram notification confirmed: '[TG] Sent OK' in serial monitor. Soil: 100% (Dry), No rain forecast, pump running.

ThingsBoard Telemetry Validation

All eight telemetry keys were received and displayed correctly on the ThingsBoard dashboard. The time-series chart showed a clear moisture transition when the potentiometer was adjusted, equivalent to the spike observed in real soil after an irrigation event.

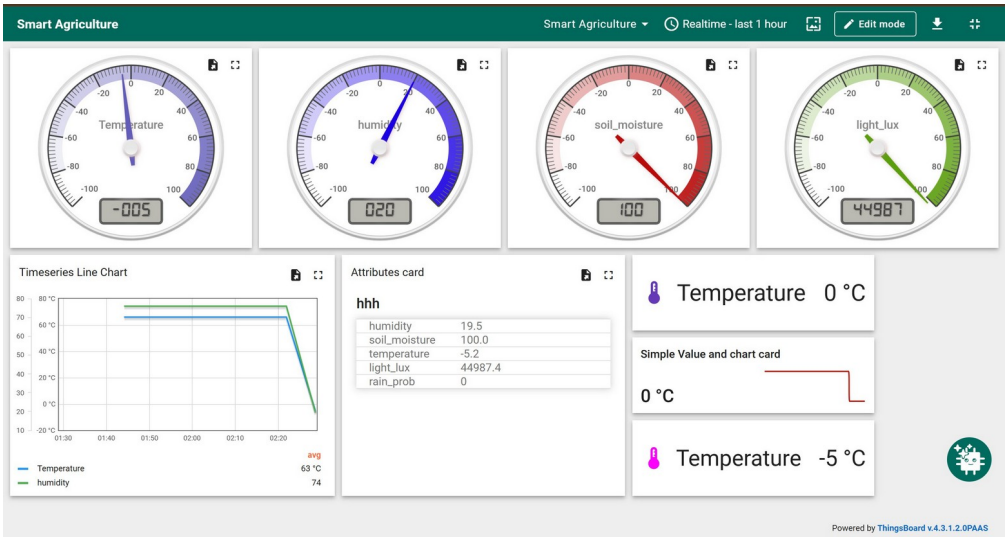

Figure 6. ThingsBoard dashboard during testing: gauges, time-series chart with visible soil moisture change event, and attributes table confirming pump_active and rain_prob values.

Results Summary

Table 2. Test Scenario Results

Test Scenario	Expected Outcome	Result
Normal (wet soil)	Pump OFF, data published	PASS
Dry soil, no rain forecast	Pump ON, Telegram alert sent	PASS
Dry soil, rain forecast ON	Pump OFF, skip alert sent	PASS
Low light	Grow LED activates	PASS
MQTT transmission	All 8 keys in ThingsBoard	PASS
Telegram delivery	[TG] Sent OK in serial monitor	PASS

8. Power Budget Analysis

Table 3. Estimated Power Consumption

Component	Current (approx.)	Notes
ESP32 (Wi-Fi + processing)	180–250 mA	Peak during MQTT TX
DHT22 + LDR + DS1307	~5 mA	Combined sensor load
Relay (coil energised)	60–90 mA	Active only when pumping
LCD2004 (backlight on)	20–40 mA	Dominant when relay off
Total (active mode)	**~260–390 mA**	Typical monitoring state
Deep Sleep (proposed)	10–50 µA	Major energy saving

On a 3,000 mAh battery in continuous mode, autonomy is approximately 8–10 hours. Implementing a 5-minute duty cycle with 15-second active periods reduces

average draw to ~15–20 mA, extending runtime to 6–10 days. A 6 W solar panel with MPPT charging would provide full energy autonomy under typical Kazakhstan summer conditions (~5.5 peak sun-hours/day).

9. Scalability Analysis

The current single-node prototype can scale in three stages:

- **Short term:** Multiple ESP32 nodes on a shared Wi-Fi network, each with a unique ThingsBoard access token, publishing to a unified dashboard -viable for greenhouse-scale deployments up to ~1 ha.
- **Medium term:** Replace Wi-Fi with LoRaWAN (Semtech SX1276) for kilometre-range, low-power coverage -suitable for open field deployment across Kazakhstan's steppe regions. LoRaWAN Class A devices achieve battery lifetimes of 5–10 years (Zhang et al., 2010).
- **Long term:** TinyML crop models on ESP32 (TensorFlow Lite for Microcontrollers), weather API integration, solar-powered nodes, and satellite imagery via Farmonaut NDVI API -supporting national Digital Agriculture 2030 objectives (Baydildina et al., 2020).

10. Conclusion

AquaGuard successfully demonstrates how affordable ESP32-based IoT technology can deliver automated, data-driven irrigation management. All project objectives were met: real-time sensor telemetry is collected, transmitted to ThingsBoard, and used by a rule-based algorithm to control irrigation while conserving water through a predictive rain-forecast feature. All six test scenarios passed. The modular architecture provides a credible pathway from this single-node prototype to a full-scale LoRaWAN field deployment, directly supporting Kazakhstan's precision agriculture goals.

References

PRECISION AGRICULTURE IN THE NORTH OF KAZAKHSTAN

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